

Implanr: An Indigenous AI-Assisted Digital Platform for End-to-End Dental Implant Case Management, Supervised Clinical Training, and Long-Term Outcome Surveillance

1. Name of Innovator(s), Designation and Affiliation

Dr. Abhijit Ashok Patil, Associate Professor, Department of Prosthodontics, Bharati Vidyapeeth Dental College and Hospital, Pune.

2. Title of the Proposed Innovation

Implanr: An Indigenous AI-Assisted Digital Platform for End-to-End Dental Implant Case Management, Supervised Clinical Training, and Long-Term Outcome Surveillance

3. Background and Need Analysis

Dental implant therapy is a multi-phase, multi-year treatment, yet in Indian teaching institutions its documentation remains largely paper-based, fragmented across departments, and unstandardized. Published audits of dental record-keeping in India repeatedly show incomplete surgical records, missing radiographic baselines, and absent long-term follow-up data — making implant survival unmeasurable and medico-legal defence difficult. No existing software manages the complete academic implant workflow: planning, supervised approvals, surgery, prosthetics, and maintenance. Implanr solves this by digitizing the entire implant lifecycle with built-in supervision hierarchies, AI-based clinical decision support, and automatic outcome analytics — improving training quality, patient safety, and research-grade data capture.

4. Problem Statement which will be addressed through this Innovation and its Prevalence

In India's 300+ dental colleges, implant cases performed by postgraduate students pass through five clinical phases spanning 6–18 months, involving students, supervisors, implant in-charges, and nursing staff. Current practice relies on paper case sheets and generic practice-management software, which cannot: (a) enforce structured phase-wise approval before irreversible surgical steps; (b) track implant failures and replacements as auditable revision chains; (c) link bone-augmentation procedures to implant outcomes; (d) generate standardized lab prescriptions and medico-legal documentation; or (e) compute survival statistics across an institution. Consequently, India — one of the fastest-growing implant markets, with implant placements growing an estimated 20–30% annually — has almost no institution-level implant survival data, and academic implantology training lacks objective performance measurement. Records are frequently lost between phases, failed implants go unanalyzed, and patients transferring between clinicians lose treatment continuity. This gap affects every dental teaching hospital and, ultimately, patient safety and evidence generation for the Indian population.

5. Key Objectives of the Proposed Technology/Project

- **Digitize and standardize the complete five-phase implant workflow** (planning → surgery → stage-2 → prosthetics → maintenance) with role-based supervised approval chains (student → supervisor → implant in-charge), ensuring no clinical phase proceeds without documented senior review.
- **Establish India's first institution-level implant outcome surveillance system** — auditable failure/replacement revision tracking, augmentation-outcome linkage, and automated survival analytics — generating research-grade longitudinal data.

- **Deploy AI-based clinical decision support** that behaves as an experienced implantologist-prosthodontist mentor, grounded in contemporary scientific literature (ITI/EAO consensus, standard textbooks), assisting students with evidence-based planning, documentation, and complication management.

6. Existing Similar Technologies

Similar solutions and their limitations:

Category	Examples	Limitations
Practice-management software	Dentrix, CareStack, Curve Dental, Open Dental	Billing/appointment-centric; no implant phase workflow, no supervised approvals, no survival tracking; costly foreign subscriptions
Implant-planning software	coDiagnostiX, DTX Studio, NobelClinician, Simplant	Cover only pre-surgical CBCT planning of a single visit; no longitudinal case management, prosthetic workflow, or training hierarchy
Dental-education logbooks	LIFTUPP (UK), generic e-logbooks	Record attendance/competency scores only; no clinical decision support, no implant-specific data, no patient-record continuity

None combines **academic supervision + full implant lifecycle + outcome surveillance + AI mentorship** in one platform, and none is priced or designed for Indian institutions.

Indigenous vs. New Innovation: Implanr is an **indigenous innovation**, conceived and developed in India within a working dental teaching hospital environment. It is directly relevant to the local market: it maps to the Indian postgraduate training structure (MDS supervision hierarchy), Indian documentation and consent norms, and the price sensitivity of Indian institutions — while being globally novel, since no international product addresses the supervised academic implant workflow end-to-end.

7. Market (Relevant) Presence / Size

Implanr is a **working functional prototype, not yet commercially launched**; it therefore holds no current market share, and the segment it targets — academic implant workflow management — is effectively unoccupied.

Potential market: India has 300+ DCI-recognized dental colleges (~250+ with MDS Prosthodontics/Oral Surgery programs placing implants), plus 5,000+ private implant-focused clinics and hospital dental departments; expansion targets include dental schools across Asia and the Middle East. The global dental practice-management software market is valued at roughly USD 2.5–3 billion (~9–11% CAGR), and the Indian dental software segment is growing faster from a low digitization base. The global dental implant market (~USD 5–6 billion, with India among its fastest-growing regions) drives demand for outcome-quality tooling.

Competitor share: the practice-management segment is dominated by foreign products (Dentrix/Henry Schein, CareStack, Curve), none of which serves academic implantology; implant-planning software (Straumann's coDiagnostiX, Envista's DTX) holds the planning niche only. Implanr fills the unclaimed gap between these two categories.

8. Prior Art Search / Intellectual Property

A prior art search was conducted across **Google Patents, WIPO PATENTSCOPE, USPTO, and InPASS (Indian Patent Advanced Search System)** using terms including "dental implant case management system," "implant workflow software," "dental treatment monitoring platform," and "implant survival tracking." Existing patents cluster around: implant hardware and abutment designs; computer-guided surgical planning

from CBCT (e.g., patents underlying coDiagnostiX/Simplant); and generic electronic health-record methods. **No prior art was found covering** a multi-phase supervised implant workflow with hierarchical clinical approvals, auditable implant failure/replacement revision chains, augmentation-outcome linkage, and integrated AI clinical mentorship — the distinguishing features of Implanr.

IP strategy: Under Section 3(k) of the Indian Patents Act, 1970, software *per se* is not patentable in India; the appropriate protection route for this innovation is **copyright**. Accordingly, **copyright registration for Implanr has already been applied for with the Copyright Office, Government of India, and is under process**. The brand name and any future hardware-integrated embodiments (e.g., device-linked ISQ capture) will be evaluated for trademark and patent protection respectively.

9. Brief Summary of the Work Plan for the Proposed Technology/Product

Preliminary work completed (working prototype / proof of concept): A fully functional prototype has been developed and internally tested, comprising: (1) the complete five-phase workflow — registration and planning (medical risk assessment, CBCT uploads, 76-implant-system library), implant surgery (torque/ISQ capture, bone and soft-tissue augmentation module), implant survival review with failure/replacement revision tracking, stage-2 surgery, prosthetic phase with automated lab prescriptions, and follow-up/maintenance; (2) five clinical roles with enforced approval hierarchy; (3) AI clinical decision support (large-language-model based) tuned to act as an evidence-grounded implantology mentor; (4) automated case summaries, operative notes, and PDF documentation with patient-data privacy safeguards; (5) institution-level analytics including survival curves and augmentation outcome comparison; and (6) advanced workflows validated in simulated cases — mid-treatment implant additions, inter-clinician case transfers, and multi-implant patients.

Future plan of action:

- **Phase I (0–6 months):** Pilot deployment in the Department of Prosthodontics, Bharati Vidyapeeth Dental College, Pune; usability testing with postgraduate students and supervisors; data-security audit.
- **Phase II (6–12 months):** Multi-centre validation across 3–5 dental colleges; native mobile app store release (Android/iOS); integration of institutional analytics dashboards; refinement from clinician feedback.
- **Phase III (12–24 months):** Commercial launch on a subscription model for dental institutions; publication of the first multi-centre Indian implant survival dataset generated by the platform; expansion to private implant clinics and international academic markets.